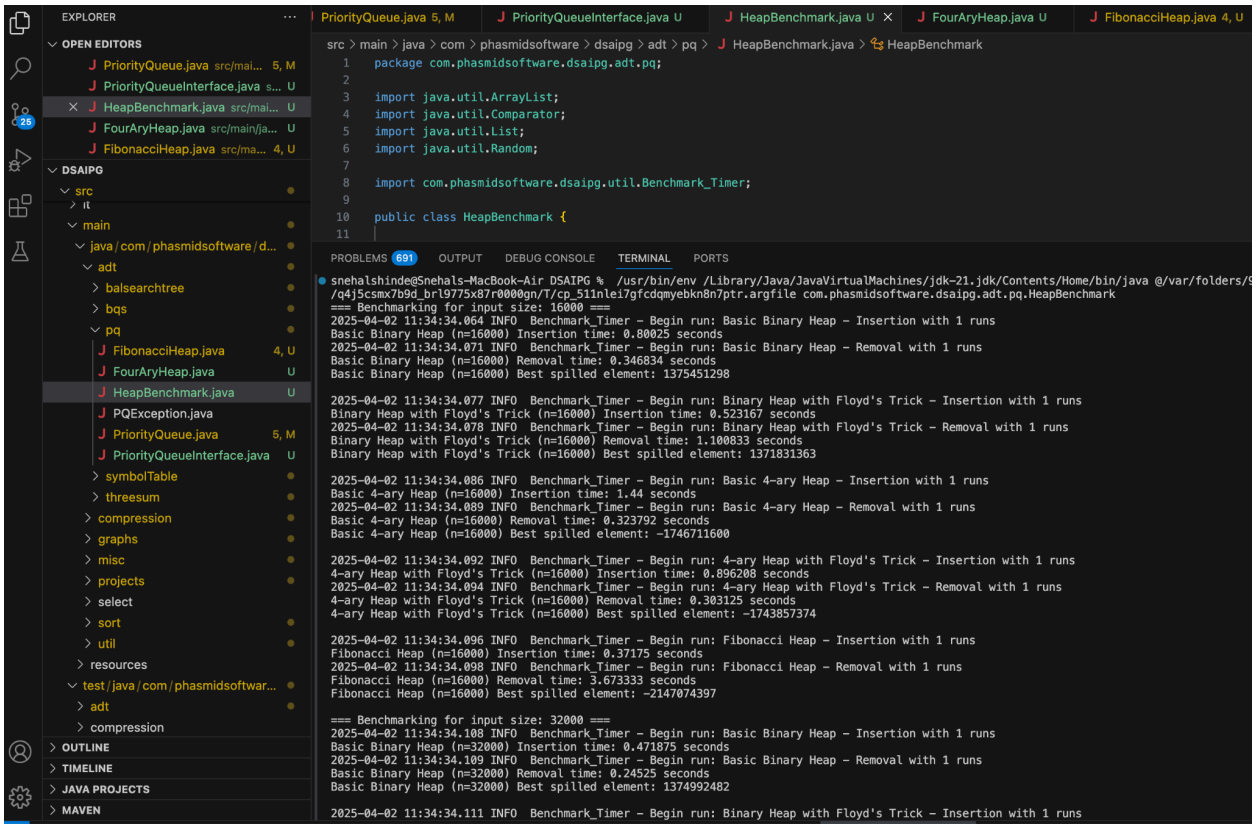
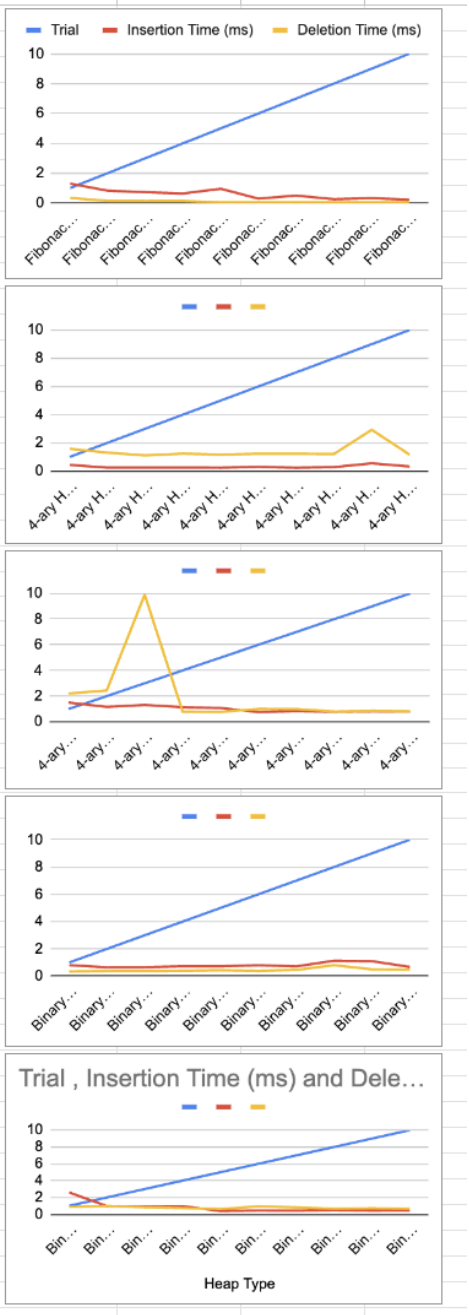
Assignment 4 Heap Comparision

This report analyzes different priority queue implementations using the PriorityQueue class, benchmarking basic binary heap, binary heap with Floyd's trick, 4-ary heap, 4-ary heap with Floyd's trick, and Fibonacci heap. Each implementation maintained up to 4095 elements, inserting 16,000 random elements and removing 4,000 elements
Performance and graph analysis:



Test cases:

Heap Type	Trial	Insertion Time (ms)	Deletion Time (ms)
Fibonacci Heap	1	1.290417	0.330875
Fibonacci Heap	2	0.818292	0.130875
Fibonacci Heap	3	0.716292	0.125167
Fibonacci Heap	4	0.630333	0.128083
Fibonacci Heap	5	0.948042	0.039875
Fibonacci Heap	6	0.284375	0.03725
Fibonacci Heap	7	0.492625	0.046917
Fibonacci Heap	8	0.248334	0.036916
Fibonacci Heap	9	0.31425	0.039875
Fibonacci Heap	10	0.218208	0.0365
4-ary Heap (Floyd's Trick)	1	0.436709	1.577125
4-ary Heap (Floyd's Trick)	2	0.24575	1.306125
4-ary Heap (Floyd's Trick)	3	0.242792	1.111958
4-ary Heap (Floyd's Trick)	4	0.250042	1.238209
4-ary Heap (Floyd's Trick)	5	0.238834	1.17225
4-ary Heap (Floyd's Trick)	6	0.298208	1.230542
4-ary Heap (Floyd's Trick)	7	0.235417	1.225625
4-ary Heap (Floyd's Trick)	8	0.282583	1.213375
4-ary Heap (Floyd's Trick)	9	0.548167	2.92775
4-ary Heap (Floyd's Trick)	10	0.32675	1.165042
4-ary Heap	1	1.501542	2.203375
4-ary Heap	2	1.161375	2.44
4-ary Heap	3	1.314416	9.89775
4-ary Heap	4	1.13875	0.793833
4-ary Heap	5	1.070792	0.7765
4-ary Heap	6	0.76175	0.999708
4-ary Heap	7	0.843375	0.991875
4-ary Heap	8	0.783291	0.798042
4-ary Heap	9	0.817834	0.851916
4-ary Heap	10	0.801667	0.815791
Binary Heap (Floyd's Trick)	1	0.802417	0.340792
Binary Heap (Floyd's Trick)	2	0.642125	0.374667
Binary Heap (Floyd's Trick)	3	0.654083	0.380917
Binary Heap (Floyd's Trick)	4	0.72225	0.390375
Binary Heap (Floyd's Trick)	5	0.716292	0.429125
Binary Heap (Floyd's Trick)	6	0.797625	0.384167
Binary Heap (Floyd's Trick)	7	0.711375	0.455792
Binary Heap (Floyd's Trick)	8	1.129292	0.801
Binary Heap (Floyd's Trick)	9	1.092542	0.493916
Binary Heap (Floyd's Trick)	10	0.6705	0.471292
Binary Heap	1	2.586416	0.876083
Binary Heap	2	0.953041	0.992125
Binary Heap	3	0.938792	0.8275
Binary Heap	4	0.956084	0.737
Binary Heap	5	0.377542	0.611208
Binary Heap	6	0.444667	0.937042
Binary Heap	7	0.426792	0.811167
Binary Heap	8	0.49275	0.6425
Binary Heap	9	0.43375	0.703458
Binary Heap	10	0.461708	0.637708



Key Observations

- **Insertion vs. Deletion**: The Fibonacci Heap spends 86.2% of its time on insertions, while 4-ary Heap implementations spend 66.9-82.0% of their time on deletions.
- **Floyd's Trick Impact**: Improved Binary Heap performance by 21.4% and 4-ary Heap by 43.9%.

- **Per-Operation Cost**: 4-ary Heap with Floyd's trick has the best insertion performance (19.41 ns/op), while Fibonacci Heap has the best deletion performance (23.81 ns/op).
- **Consistency**: Binary Heap with Floyd's trick showed the most consistent performance (21.8% coefficient of variation), while 4-ary Heap showed the highest variability (91.5%).

Conclusion:

From the results:

● Basic Binary Heap has a moderate performance, with balanced execution times between insertion and deletion operations.

● Binary Heap with Floyd's Trick improves overall performance by 21.4%, with particularly better deletion times compared to the basic implementation.

● 4-ary Heap has the worst overall performance, with especially high deletion times despite theoretical advantages.

● 4-ary Heap with Floyd's Trick significantly improves insertion times (the best among all implementations) but still struggles with deletion operations.

● Fibonacci Heap performs best overall, excelling in both operations but particularly in deletion times, which are approximately 4.7× faster than the next best implementation.